

# SmartDialer System Architecture

## Engineering Design, Concurrency Models, Safety Bounding & State Machines

### 1. Executive Summary & Core Principle

SmartDialer is a production-grade functional prototype of an outbound progressive and predictive dialing system tailored for debt collection and high-touch contact environments. The core engineering mandate is strict concurrency correctness, safety bounding, idempotency, and graceful degradation without introducing unnecessary distributed complexity such as Kafka, Redis, or microservices.

**CORE ARCHITECTURAL INVARIANT:** Prediction Proposes → Safety Decides → Allocation Executes → Carrier Emits Events.

The Predictive Pacing Engine holds ZERO references to Telecom Providers or Call Allocators. The Safety Controller is the final authority that guarantees zero borrower abandonment by calculating real-time capacity bounds before any call can be initiated.

### 2. Linear Decision & Event Ingestion Pipeline

| Pipeline Stage | Component        | Responsibility & Data Flow                                                                                                           |
|----------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| 1. Proposal    | PredictiveEngine | Computes suggested call volume using Exponential Moving Average (EMA) of answer rates and talk duration. Outputs an integer request. |
| 2. Bounding    | SafetyController | Inspects live available agents, in-flight limits, provider health, and circuit breaker state. Issues APPROVE, REDUCE, or REJECT.     |
| 3. Execution   | CallAllocator    | Atomically reserves agent and borrower with a lease deadline, creates Call record in RESERVED state, and calls provider.             |
| 4. Telephony   | TelecomProvider  | Provider A (clean) / Provider B (chaotic). Delivers async telephony signaling events via daemon threads with unique event IDs.       |
| 5. Ingestion   | EventProcessor   | Ingests webhooks, enforces O(1) deduplication, applies monotonic rank state transitions, and releases completed agents.              |
| 6. Recovery    | Reconciler       | Out-of-band periodic scan that recovers orphaned agents and stuck calls from crashed workers using lease deadlines.                  |

### 3. Component Deep Dive

#### 3.1 In-Memory Repository with PostgreSQL Migration Path (StateStore)

StateStore uses a two-level locking model: entity-level locks for snapshot reads, and per-row locks for atomic mutations. Every method directly mirrors single SQL statement semantics. For example, `atomic_reserve_agent` performs an atomic check-and-set directly equivalent to: `UPDATE agents SET state='RESERVED', reservation_id=:res, lease_until=:lease WHERE id=:id AND state='AVAILABLE';`.

#### 3.2 Circuit Breaker Pattern (CircuitBreaker)

Tracks carrier health across 3 states: **CLOSED** (normal traffic), **OPEN** (consecutive failures  $\geq$  threshold; all new calls blocked), and **HALF\_OPEN** (cooldown elapsed; exactly 1 atomic probe call allowed). The Safety Controller falls back to safe single-call progressive mode when **HALF\_OPEN**.

### 3.3 Idempotent Event Processing (EventProcessor)

Each call record maintains a processed\_event\_ids set. Incoming events check set membership in O(1). Duplicate webhook deliveries are dropped immediately before triggering any state transitions or version increments.

### 3.4 Lease-Based Worker Crash Recovery (Reconciler)

When an allocation occurs, a timestamp lease (lease\_until = now + N) is written to the Call and Agent. If a worker process crashes mid-reservation, the Reconciler discovers the expired lease and transitions pre-provider calls to CANCELLED and post-provider calls to FAILED. **Crucially, active live calls (CONNECTED) are never terminated.**

## 4. State Machine Formalisms

| State Entity    | Valid States                                                                                                       | Transition & Ordering Invariants                                                                                         |
|-----------------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Agent           | OFFLINE, AVAILABLE, RESERVED, DIALING, CONNECTED, WRAP_UP, PAUSED                                                  | Atomic reservation prevents double-booking. Wrap-up automatically frees agent to AVAILABLE.                              |
| Call            | QUEUED(0), RESERVED(1), INITIATED(2), RINGING(3), ANSWERED(4), CONNECTED(5), COMPLETED(6), FAILED(6), CANCELLED(6) | Monotonic rank ordering. Backwards transitions rejected. Terminal states (COMPLETED, FAILED, CANCELLED) are black holes. |
| Circuit Breaker | CLOSED, OPEN, HALF_OPEN                                                                                            | Threshold trips to OPEN. Cooldown timer enables HALF_OPEN probe. Probe success resets to CLOSED.                         |